CISC 473: Deep Learning — Midterm Report

P02: Noise-Aware Zero-Reference Low-Light Image Enhancement

Group Members:

Bella Xu 20351957 (Role: Project Lead and Research Lead)
Mark Nistor 20412135 (Role: Engineering and Research Lead)

Two-member group as approved by the instructor.

GitHub: https://github.com/MarkLaMer/Noise-Aware-Deep-Curve-Estimation

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Abstract

Low-light images often suffer from reduced visibility, colour distortion, and amplified sensor noise, limiting both visual quality and downstream computer vision performance. Zero-Reference Deep Curve Estimation (Zero-DCE) enhances images using learned pixel-wise adjustment curves and non-reference training losses, avoiding the need for paired normal-light targets. However, its training objective does not explicitly address noise amplification in dark regions.

Our project investigates whether lightweight noise-aware regularisation can reduce this problem while preserving the zero-reference training framework. We reproduce and evaluate Zero-DCE on the LOL v1 dataset to establish a baseline for investigating noise amplification. Our reproduced model achieves a PSNR of 14.96 ± 0.41 dB and an SSIM of 0.516 ± 0.003 across three random seeds, compared with 14.86 dB and 0.562 for the official checkpoint. We observe scene-dependent enhancement behaviour, including over-exposure in some dark scenes. Future work will investigate lightweight noise-aware regularisation, with Zero-DCE++ reproduction considered if time permits. Three potential extensions are considered: darkness-gated smoothness, self-supervised neighbouring-pixel consistency, and curve-gain regularisation.

The remaining work will focus on investigating these limitations and implementing a targeted noise-aware regularisation method without increasing model complexity.

Introduction and Problem Statement

Images captured under poor lighting conditions often suffer from low contrast, colour distortion, and significant sensor noise. These issues not only reduce image quality for human viewing but also affect downstream computer vision tasks, including object detection, image classification, and semantic segmentation.

Traditional supervised low-light enhancement methods rely on paired low-light and normal-light images for training. However, collecting such datasets can be challenging because lighting conditions and scene content may change between captures. Zero-Reference Deep Curve Estimation (Zero-DCE) addresses this limitation by learning image-specific enhancement curves using non-reference loss functions. Zero-DCE++ further reduces computational complexity while maintaining the zero-reference approach.

Despite these advantages, neither method explicitly models sensor noise. When dark regions are brightened, existing noise can become more visible, potentially reducing the quality of the enhanced

image.

Our research question is:

Can noise amplification in Zero-DCE-style enhancement be reduced through lightweight, noise-aware regularisation without requiring clean reference images during training or introducing additional denoising networks?

To investigate this question, we have reproduced and evaluated the Zero-DCE baseline and identified scene-dependent enhancement behaviour. Our next step is to investigate lightweight noise-aware regularisation methods to reduce noise amplification.

Since the original proposal, our research question has remained unchanged, but we have narrowed our focus to reproducing Zero-DCE and developing a noise-aware regularisation method. We have completed baseline training and evaluation, and will next focus on implementing and testing noise reduction. Zero-DCE++ reproduction and additional evaluations will be considered if time permits.

Following the course protocol, our baseline was trained on LOL low-light images rather than the SICE dataset used for the official Zero-DCE model.

Method	**Year**	**Key idea**	**Relation to our work**
Zero-DCE	2020	Zero-reference curve estimation	Main baseline; lacks explicit noise suppression
Zero-DCE++	2021	Lightweight zero-reference enhancement	Efficiency and image-quality baseline
MIRNet	2020	Multi-resolution supervised restoration	Quality reference; requires paired training data
SNR-Aware	2022	SNR-guided low-light enhancement	Demonstrates benefits of noise-aware processing
ZLEN	2024	Noise estimation with zero-reference enhancement	Motivates noise-aware regularisation
ZERO-IG	2024	Joint denoising and illumination-guided enhancement	Related approach using additional components
Zero-DiDCE	2024	Adaptive enhancement curves	Highlights noise amplification issues
RUAS	2021	Retinex-inspired unrolling with reference-free architecture search	Alternative lightweight enhancement approach without reference images
SCI [11]	2022	Self-calibrated illumination learning with unsupervised loss	Demonstrates efficient enhancement without paired training targets

Updated Related Work

Recent studies have explored different approaches to low-light image enhancement. Wei et al. [12] introduced Retinex-based decomposition, while Liu et al. [10] developed a Retinex-inspired unrolling framework. Ma et al. [11] focused on fast and flexible enhancement, and Jiang et al. [13] explored wavelet-based diffusion models. Li et al. [14] provided a broader survey of deep learning approaches.

Building on these studies, our work focuses on reducing noise amplification in Zero-DCE through lightweight noise-aware regularisation without requiring paired training images. Neighbor2Neighbor [9]

provides a self-supervised denoising approach without clean reference images, motivating our investigation of noise-aware regularisation.

Methodology

Model Architecture

We implemented the original Zero-DCE baseline from scratch using PyTorch. The model employs a seven-layer convolutional network (DCE-Net) to estimate pixel-wise enhancement parameters.

All convolutional layers use 3×3 kernels with stride 1 and padding 1. The first six layers use ReLU activation, while the final layer uses Tanh to constrain the predicted curve parameters to $[-1, 1]$. Skip connections are implemented by concatenating the outputs of layers 3 and 4 as input to layer 5, layers 2 and 5 as input to layer 6, and layers 1 and 6 as input to layer 7, pairing earlier layers symmetrically with later ones. Concatenation is performed along the channel dimension, increasing the input to layers 5–7 from 32 to 64 channels (32 + 32). These connections preserve fine spatial detail from the earlier layers and pass it directly to the later layers, helping retain image structure. Since no layer downsamples (stride 1 and padding 1 throughout), the network predicts curve parameters at full resolution, giving every pixel its own enhancement curve.

The network contains 79,416 trainable parameters and produces 24 output channels, corresponding to eight sets of RGB enhancement curves.

The enhancement operation is defined as:

$$I_{n+1}(x) = I_n(x) + A_n(x) I_n(x)(1 - I_n(x)), \tag{1}$$

where $I_n(x)$ is the pixel intensity after iteration n, and $A_n(x)$ is the estimated curve parameter. The transformation is applied for eight iterations. Our implementation follows the curve transformation defined in the original paper, $I + AI(1 - I)$, while the official implementation expresses the same transformation as $I + r(I^2 - I)$, where $r = -A$. Therefore, when evaluating the released official checkpoint, the predicted curve maps are negated to account for this difference in sign convention.

The curve parameters A are restricted to the range $[-1, 1]$ using a tanh activation function. This ensures that the curve remains monotonically increasing, since its derivative with respect to I, $1 + A(1 - 2I)$, is nonnegative for $I \in [0, 1]$ whenever $|A| \leq 1$. Together with the fixed endpoints (the curve maps 0 to 0 and 1 to 1 for any A), this preserves the brightness order of pixels and keeps the enhanced intensities within $[0, 1]$.

The model is initialised with random weights rather than pretrained parameters. Specifically, convolutional weights are sampled from a Gaussian distribution with a mean of 0 and a standard deviation of 0.02 (i.e., $\mathcal{N}(0, 0.02^2)$), following the official implementation, while biases retain PyTorch's default initialisation.

Our planned noise-aware modifications target the training loss rather than the network architecture.

Training Setup

The baseline training pipeline uses the Adam optimiser with a learning rate of 10^{-4}.

The total training objective is a weighted sum of four non-reference losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{spa}} + 10\,\mathcal{L}_{\text{exp}} + 5\,\mathcal{L}_{\text{col}} + 200\,\mathcal{L}_{\text{tv}}. \tag{2}$$

Throughout, the brightness of a pixel is the mean of its R, G and B values.

Spatial consistency loss. This loss preserves the relative intensity differences between neighbouring regions in the input image X and the enhanced image Y:

$$\mathcal{L}_{\text{spa}} = \frac{1}{N}\sum_{p=1}^{N}\sum_{q\in\Omega(p)}\Big[(\bar{Y}_p - \bar{Y}_q) - (\bar{X}_p - \bar{X}_q)\Big]^2, \tag{3}$$

where $\bar{X}_p$ and $\bar{Y}_p$ are the mean brightness of the 4×4 region p in the input and enhanced images, $\Omega(p)$ is the set of its four neighbouring regions (left, right, up, down; regions outside the image are treated as zero), and N is the number of regions over which the loss is averaged.

Exposure control loss. This loss encourages each enhanced-image patch to have a mean intensity near the target exposure level E:

$$\mathcal{L}_{\text{exp}} = \frac{1}{M}\sum_{k=1}^{M}\big(\tilde{Y}_k - E\big)^2, \tag{4}$$

where $\tilde{Y}_k$ is the mean brightness of the k-th non-overlapping 16×16 patch of the enhanced image, M is the total number of patches, and $E = 0.6$ for images normalised to $[0,1]$.

Colour constancy loss. This loss discourages differences between the spatially averaged red, green and blue channel intensities of each enhanced image:

$$\mathcal{L}_{\text{col}} = \frac{1}{B}\sum_{b=1}^{B}\sqrt{(\mu_{R,b}-\mu_{G,b})^4 + (\mu_{R,b}-\mu_{B,b})^4 + (\mu_{G,b}-\mu_{B,b})^4}, \tag{5}$$

where B is the batch size and $\mu_{R,b}$, $\mu_{G,b}$ and $\mu_{B,b}$ are the channel means of image b.

Illumination smoothness loss. This loss penalises abrupt horizontal and vertical changes in the predicted curve-parameter maps A:

$$\begin{aligned}
\mathcal{L}_{\text{tv}} &= \frac{2}{B}\left[\frac{S_v}{(H-1)W} + \frac{S_h}{H(W-1)}\right],\\
S_v &= \sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{i=1}^{H-1}\sum_{j=1}^{W}\big(A_{b,c,i+1,j} - A_{b,c,i,j}\big)^2,\\
S_h &= \sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W-1}\big(A_{b,c,i,j+1} - A_{b,c,i,j}\big)^2,
\end{aligned} \tag{6}$$

where H and W are the image height and width and $C = 24$ is the number of curve-parameter maps per image (8 iterations $\times$ 3 colour channels).

Together, these losses encourage the preservation of local structure, suitable exposure, balanced colours and spatially smooth curve parameters, without requiring paired ground-truth images.

The loss functions and their weighting coefficients follow the authors' official implementation, with weights of 1, 10, 5 and 200 for spatial consistency, exposure control, colour constancy and illumination smoothness, respectively. These differ from the original paper, which specifies weights of 0.5 for colour constancy and 20 for illumination smoothness, while leaving the spatial consistency and exposure terms unweighted.

The mathematical formulations also differ. For example, the paper defines exposure loss using absolute differences, whereas the official code uses squared differences. Similarly, the paper defines colour constancy loss as a sum of squared channel differences, while the code uses the square root of the sum of their fourth powers. The official implementation was followed to maintain consistency with the authors' released model and to support reproducibility when comparing against the official checkpoint in Table 2.

Hyperparameter	**Value**
Optimiser	Adam
Learning rate	1×10^{-4}
Batch size	8
Epochs	100
Exposure target E	0.6
Training resolution	256×256
Test resolution	600×400 (no resize)
Weight decay	1×10^{-4}
Gradient clipping	0.1
Loss function	Four non-reference losses
Data augmentation	None implemented
Random seeds	1, 2, 3
Hardware	NVIDIA RTX 4070 Laptop GPU
Training time	Approximately 10 min per run
Framework	PyTorch 2.14.1
Checkpoint used	Final epoch (100)

Table 1: Training configuration shared by all three runs (seeds 1, 2 and 3).

Dataset and Preprocessing

The model was trained on the 485 low-light images from the LOL dataset's training split (`our485/low`) [12]. During preprocessing, images were converted to three-channel RGB, resized to 256×256 pixels using Lanczos interpolation, and normalised to the range $[0, 1]$. The images were loaded in shuffled mini-batches, with only the low-light images used for training, preserving the zero-reference setting.

The implementation did not include random data augmentation or an explicit validation split. Evaluation was performed using the 15 image pairs from the LOL `eval15` test split at their original resolution of 600×400 pixels, without resizing. The corresponding normal-light reference images were used exclusively for computing evaluation metrics and were never used during training.

Preliminary Results

Evaluation Protocol

Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) were computed using the scikit-image library at the original image resolution. The enhanced outputs were clamped to $[0, 1]$ and quantised to 8-bit RGB images before evaluation, with SSIM computed across the colour channels.

The same evaluation procedure was applied to both our trained models and the official Zero-DCE checkpoint, reproducing the published PSNR of 14.86 dB and SSIM of 0.562 [13]. Reported SSIM values for Zero-DCE vary from approximately 0.54 to 0.589 across publications [14, 13], potentially due to differences in evaluation procedures and metric implementations. Evaluating the official checkpoint using the same procedure therefore provides a consistent and fair basis for comparison with our trained models.

Baseline Reproduction

Model	**PSNR (dB)**	**SSIM**
Zero-DCE, published	14.86	0.562
Zero-DCE, official checkpoint (measured by our evaluator)	14.86	0.562
Ours, LOL-trained (3 seeds)	14.96 ± 0.41	0.516 ± 0.003

Table 2: Preliminary PSNR and SSIM results on the LOL v1 eval15 test split (15 image pairs, full resolution). Our row is mean ± sample std over 3 seeds; the two Zero-DCE rows are single models.

As shown in Figure 2, the total training loss for seed 1 decreased from 2.7135 at epoch 1 to 0.6677 at epoch 100, representing a reduction of approximately 75.4%. The minimum loss of 0.6633 was recorded at epoch 96.

Qualitative Analysis

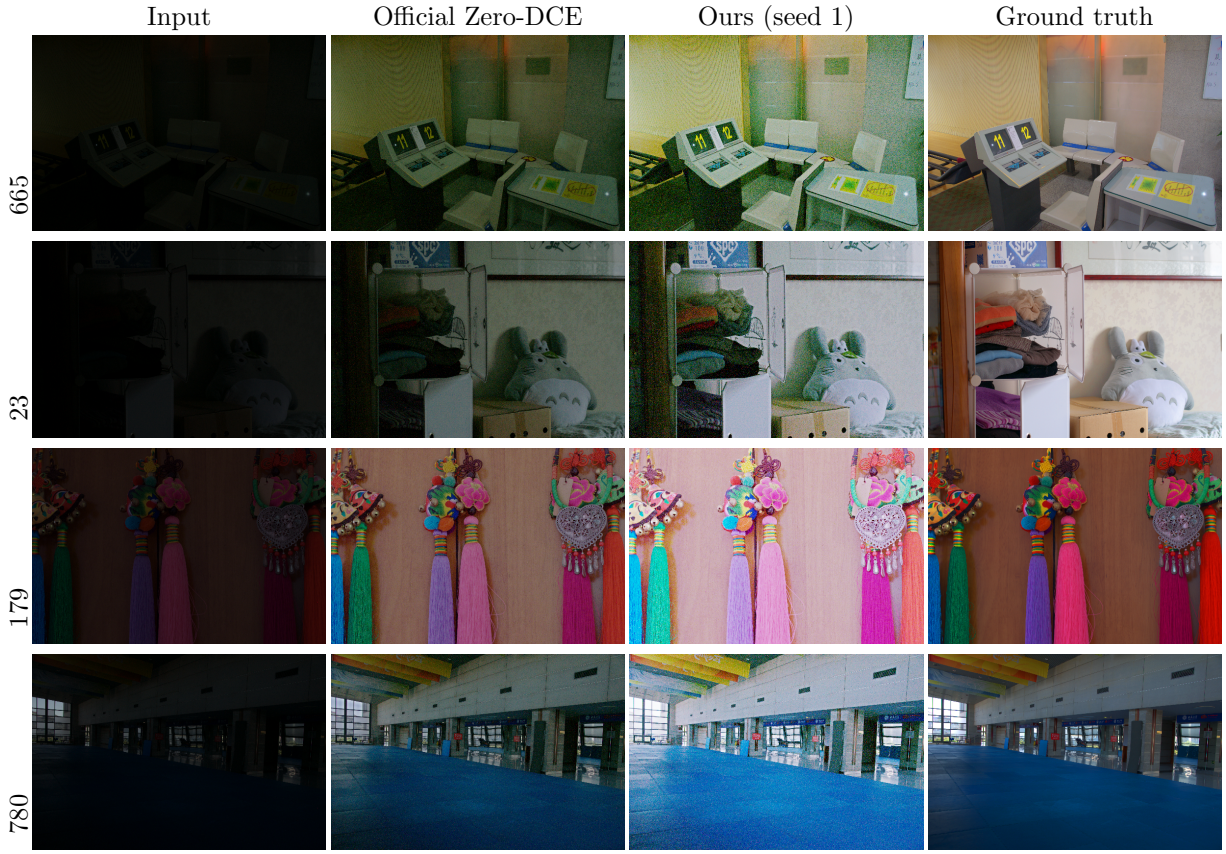


Figure 1: Full-resolution qualitative comparison on four LOL eval15 test images (IDs 665, 23, 179, 780). Columns show the dark input, official Zero-DCE (released checkpoint), ours (seed 1), and ground truth, from left to right. For images 665 and 23, the official model under-brightens while ours better matches the ground-truth brightness (PSNR: 10.8/18.9 and 8.3/16.4 dB, official/ours); for images 179 and 780, ours over-exposes scenes whose ground truth is dim (PSNR: 20.2/7.8 and 22.5/11.1 dB). Noise is visible in dark regions of both models' enhanced outputs, particularly on the wall in image 665.

Discussion

The experimental results demonstrate that the Zero-DCE architecture was successfully reproduced, with training converging consistently across the three random seeds (final training loss 0.668 ± 0.001, with negligible change over the last 30 epochs). The evaluation procedure reproduced the published results of the official checkpoint exactly, confirming that our PSNR and SSIM implementation matches the one used in the literature. As shown in Table 2, our trained model achieved a PSNR approximately 0.7% higher than the official model, meeting the reproduction target. However, this difference was smaller than the variation observed across random seeds and therefore indicates comparable performance rather than a meaningful improvement. In contrast, SSIM was approximately 8.3% lower, suggesting that the reproduced model preserved structural similarity less effectively despite achieving similar pixel-level reconstruction quality; the over-exposure discussed below is a likely contributor, since it reduces local contrast, which SSIM penalises.

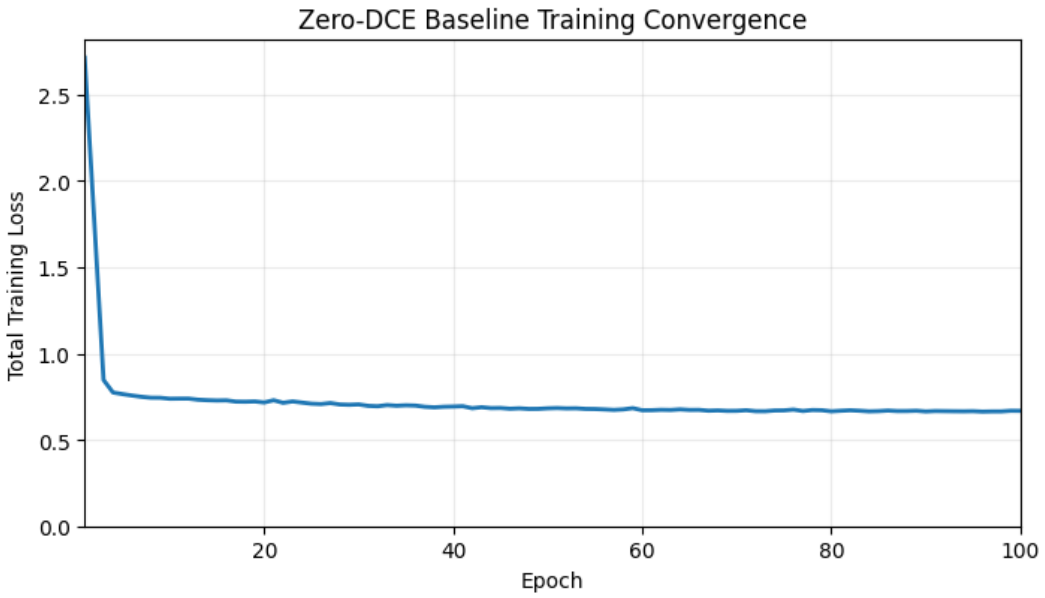


Figure 2: Total training loss of the reproduced Zero-DCE baseline over 100 epochs (seed 1).

Qualitative comparisons (Figure 1) revealed that the similar mean PSNR arises from opposite behaviours. The official checkpoint tended to produce darker outputs, while our trained model applied stronger illumination adjustments, pushing image brightness toward the target exposure level of $E = 0.6$. For brighter scenes such as images 665 and 23, this favoured our model (PSNR 10.8 to 18.9 dB and 8.3 to 16.4 dB, official to ours), whereas for genuinely dim scenes such as images 179 and 780 our model over-exposed the result (PSNR 20.2 to 7.8 dB and 22.5 to 11.1 dB). Two factors likely explain this. First, the models were trained on different data: the official checkpoint on SICE Part 1, and ours on LOL low-light images following the course protocol, which influences the learned enhancement curves. Second, the exposure control loss pushes every scene toward the same fixed target regardless of how bright it should appear, so scenes whose ground truth is dim are over-brightened. At the final epoch of seed 1, the weighted exposure loss accounted for approximately 64.6% of the total training loss. This indicates that exposure control remained a substantial component of the training objective, although its relative magnitude does not establish its causal contribution to over-exposure.

A limitation observed in the enhanced images was the amplification of visible noise in darker regions. While the exposure control loss encourages increased brightness, the loss function does not explicitly penalise image noise. Consequently, brightening poorly illuminated regions can also increase the visibility of existing sensor noise. The illumination smoothness loss encourages spatially consistent curve parameters but does not directly remove noise from the enhanced image. This limitation motivates the work planned for Week 7, which will investigate approaches to reducing noise amplification while maintaining effective low-light enhancement.

Challenges and Adaptations

One of the main challenges was the late start to the project, which reduced the time available for initial implementation and experimentation. Training was initially unexpectedly slow because the installed PyTorch version did not support CUDA acceleration. After switching to a CUDA-

enabled installation and using the NVIDIA RTX 4070 Laptop GPU, training time decreased to approximately 10 minutes per run, allowing us to complete experiments across three seeds.

Another challenge was identifying discrepancies between the original Zero-DCE paper and the authors' official implementation. Several loss functions differed in their mathematical formulations and weighting coefficients, particularly the exposure, colour constancy and illumination smoothness losses. Additionally, the official implementation used a different sign convention for the enhancement curves. These differences initially complicated reproduction of the published method. To address this, the official source code was examined and its loss formulations and weighting coefficients were adopted. The curve sign convention was also accounted for when evaluating the official pretrained checkpoint.

A further challenge was ensuring that the evaluation procedure produced reliable and comparable results. Differences in image preprocessing and metric implementations can affect PSNR and SSIM scores, making direct comparisons with published results difficult. To validate our evaluation procedure, the official Zero-DCE checkpoint was evaluated on the LOL test dataset using the same evaluator as our trained models. This reproduced the published PSNR of 14.86 dB and SSIM of 0.562, providing confidence in the evaluation pipeline and allowing a consistent comparison between the official checkpoint and our reproduced models.

Although the late start and initial training issues delayed progress, switching to GPU training allowed the baseline experiments and evaluation to be completed. The remaining project timeline has been adjusted to focus on analysing the observed limitations and improving enhancement quality.

Updated Week 7–10 Plan

Week	Tasks	Responsible
7	Implement and test the noise-aware regularisation losses (darkness-gated smoothness, neighbour consistency, curve-gain); conduct initial experiments and document training behaviour.	Mark
8	Ablations of the loss variants and their weights, each trained with 3 seeds (mean ± std); compare against the baseline using PSNR, SSIM and the noise study.	Mark
9	Evaluate enhancement quality using PSNR, SSIM, LPIPS, and NIQE; perform feasible ablation studies and analyse qualitative results.	Bella & Mark
10	Finalise experimental results, complete the final report, prepare presentation slides, and review the project deliverables. **(Final due)**	Bella & Mark

Noise Amplification Analysis (Preliminary Novel Contribution)

To quantify noise amplification, each of the 15 LOL `eval15` images was divided into 20×20 patches, and the 30% least textured patches (lowest Sobel gradient on the normal-light reference) were kept, giving 2,700 flat patches in which pixel variation reflects noise rather than image detail. Noise was estimated per patch as the standard deviation of the residual after subtracting a Gaussian-blurred ($\sigma = 2$) version of the image. Patches were split into five equal-size bins by their brightness in

the low-light input. *Noise gain* is the ratio of output to input noise, and *excess noise* the ratio of output to reference noise; values above 1 mean the enhanced image is noisier.

Bin	Input brightness	Patches	Noise gain (ours)	Excess (ours)	Excess (official)
1 (darkest)	0.001–0.024	540	21.5	8.8	3.3
2	0.024–0.050	540	10.0	7.8	4.5
3	0.050–0.070	540	5.3	5.6	4.6
4	0.070–0.102	540	3.6	3.8	3.7
5 (brightest)	0.102–0.309	540	1.7	2.3	3.2

Table 3: Noise amplification by input-brightness bin for our model (seed 1) and the official Zero-DCE checkpoint on LOL `eval15`. Brightness is normalised to $[0, 1]$; noise gain is relative to the low-light input, excess noise relative to the normal-light reference.

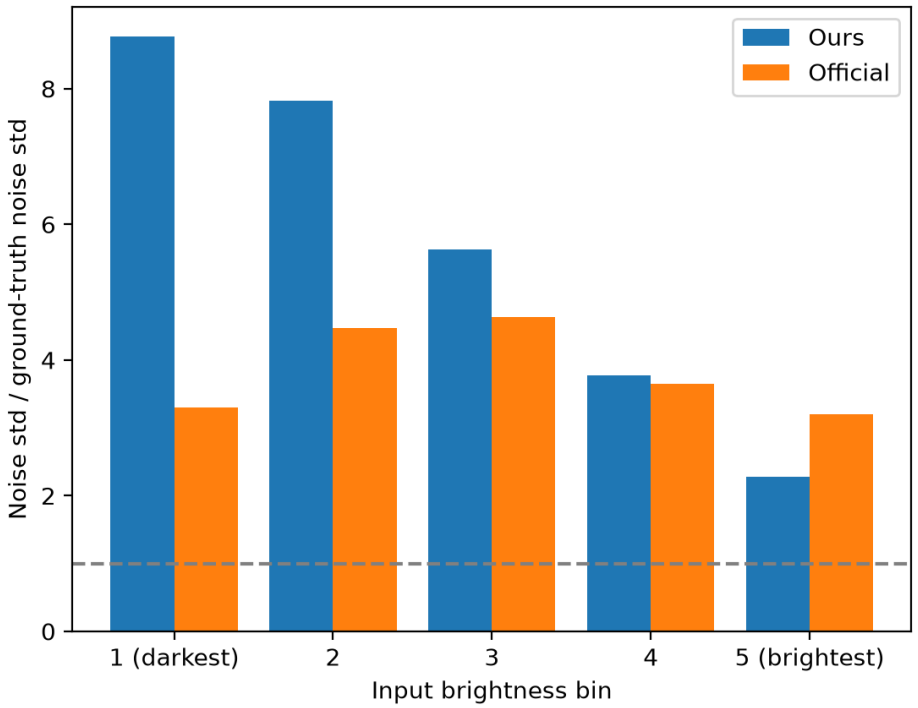


Figure 3: Excess noise (output noise divided by reference noise) per input-brightness bin, darkest on the left. The dashed line marks the reference noise level.

Both models are noisier than the reference in every bin (Table 3, Figure 3). For our model, excess noise rises steadily as the input gets darker, from 2.3 in the brightest bin to 8.8 in the darkest, whereas the official checkpoint stays between 3.2 and 4.6. Our model is therefore about 2.7 times noisier than the official one in the darkest regions, yet slightly cleaner in the brightest, consistent with its stronger enhancement of dark inputs. The raw noise gain of 21.5 in the darkest bin is partly inflated, since these inputs span only a few 8-bit levels and their measured noise is very small; excess noise is the more reliable measure. Overall, these results quantify a limitation of the Zero-DCE loss formulation, which does not penalise noise, and motivate the noise-aware losses planned for Week 7.

AI Usage

AI tools, primarily Claude, were used throughout the project as a supporting resource for research, debugging and writing. This included assistance with literature searches, verifying published performance metrics and identifying differences between the Zero-DCE paper and its official implementation. AI was also used to help debug our own code and provide a tutorial-style guide for understanding and implementing the model.

Contribution Declaration

Member	Primary Responsibilities	Estimated %
Bella Xu	Submission logistics, literature review, data collection coordination, report writing, analysis	50%
Mark Nistor	Model implementation, report writing, analysis, experiments	50%

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